

Nonlinear Soliton Dynamics in Polycatenated Architected Materials (PAMs)

Background Info and Motivation

Polycatenated architected materials (PAMs) are a new class of metamaterials composed of interlocked, ring-like units whose mechanical behavior is governed by both constituent material properties as well as the unit/linkage topology. The high internal degrees of freedom from these interlocking connections give rise to nonlinear force-displacement relationships, enabling PAMs to exhibit tunable stiffness, energy dissipation, and wave propagation characteristics. Prior studies have primarily focused on the static and quasi-static mechanical response of PAMs. However, understanding their dynamic behavior (especially how mechanical impulses propagate through chains and networks of PAM units) is essential for use in applications such as shock absorption, impact mitigation, and mechanical signal control.

Theoretically, PAM chains can be modeled as masses connected by nonlinear springs. These models predict that system behavior depends critically on the ratio between impulse force and pretension:

- In the **weakly nonlinear regime** (large pretension relative to impulse), the system behaves approximately linearly, producing dispersive waves
- In the **strongly nonlinear regime** (large impulse relative to pretension), the system supports **solitons**: localized, non-dispersive, and particle-like excitations

Last quarter, we continued upgrading and refining our experimental platform to probe these dynamics but were unable to reliably observe soliton formation. This quarter, we addressed key physical issues and successfully generated soliton-like excitations consistent with theoretical predictions.

Key Points of Improvement

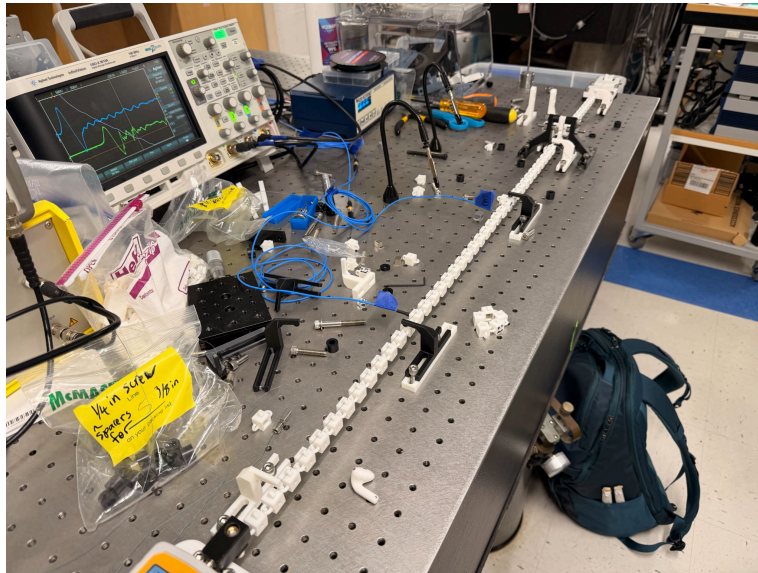


Fig 1: an image of the experimental setup, where a 1D PAM chain is suspended by strings and pretension, with 2 accelerometer blocks embedded in different places (where the blue cables go) to measure the mechanical pulse that propagates through the system. The impulse actuation site was originally at the far end, but was moved closer to the center (at around $\frac{3}{4}$ of the way down the chain) to avoid mechanical issues.

A key issue identified in our previous setup was due to how the impulse was applied at a fixed boundary. In earlier experiments, one end of the PAM chain was rigidly secured using a long screw, and actuation was applied directly at this boundary. However, this configuration introduced a significant normal reaction force during impact, which distorted the applied excitation and made it less impulse (delta function)-like. Instead of producing a sharp, localized input, the system experienced a more distributed force over time, which resulted in dispersive wave propagation rather than soliton formation. (Interestingly, earlier observations of soliton-like behavior based on manual, inconsistent actuations were likely due to the presence of a piezo actuator at the boundary, which, while no longer in use, allowed some loose motion. This slight looseness allowed motion at the boundary and happened to produce a cleaner impulse). When we removed the piezo actuator and rigidly fixed the boundary, the system consistently exhibited wave-like responses instead of localized excitations, showing that boundary compliance plays an important role.

To address this, we moved the actuation point away from the boundary and into the middle of the chain. By extending the chain length on both sides of the actuation site, we ensured that reflections from the boundaries would not interfere with the primary signal within the observation window. In addition, we replaced rigid boundary constraints with a pulley-based pretension system using hanging masses. Compared to fixed screws, this allows for more compliant boundary conditions and reduces the amplitude of reflected waves, allowing the system to better approximate an effectively infinite chain.

Another challenge involved ensuring that the applied impulse was aligned with the axis of the chain. When forces are applied off-axis, they introduce torque and transverse motion, causing motion to be diverted into lateral and rotational modes. This disrupts the formation of a clean soliton excitation. To verify this, we performed a simple test in which we struck a unit (with an extended rectangular extrusion for force application) at varying distances from the chain axis.

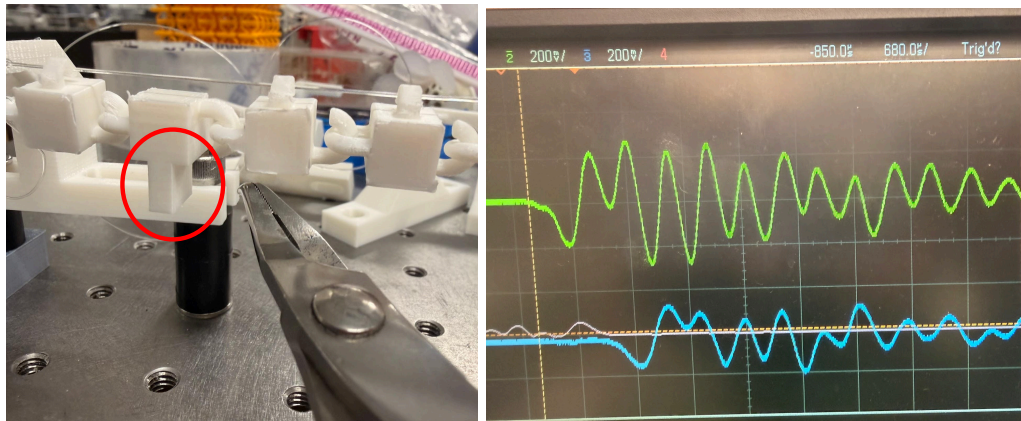


Fig. 2: Left image shows the unit with the rectangular extrusion to provide a site to apply an impulse onto that is farther away from the center of the chain axis to investigate how the waveform is affected. As expected, the resulting pulse (right image) is much more wave-like: the opposite of the soliton-like pulse we wish to investigate.

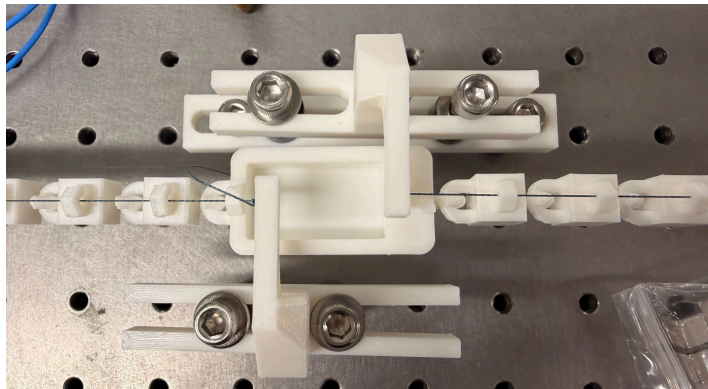


Fig 3: The striker box unit, which allows access to the center axis of the chain through one of its walls for more direct, centered impulse application.

As shown in Fig. 2, impacts applied farther from the axis produced more dispersive, wave-like responses, whereas impacts closer to the axis resulted in more localized behavior. Based on this observation, we designed a modified unit, referred to as the striker box (Fig. 3), which features an open central cavity that allows direct access to the chain axis. This ensures that the applied force is aligned with the longitudinal center axis of the chain.

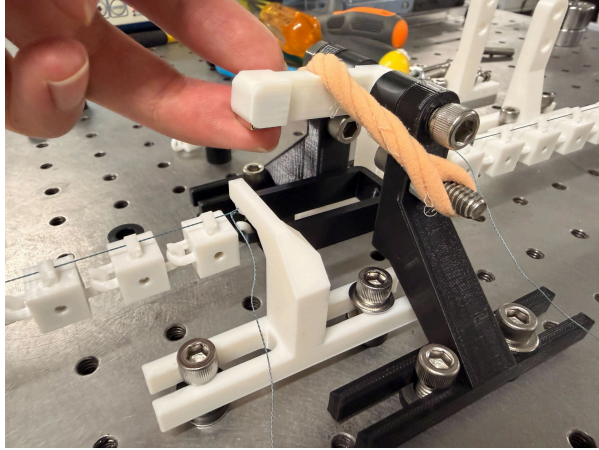


Fig. 4: The final striker mechanism, utilizing a small swinging hammer-like striker with the same metal block embedded in the tip as the PAM units to match masses. It swings down accelerated by gravity and/or the elastic band to hit the black striker box at the center of the chain axis.

We also considered the role of momentum transfer in generating a clean impulse. From conservation of momentum, if the striker and the PAM unit have significantly different masses, the resulting collision leads to an uneven distribution of momentum and a less localized excitation. To address this, we increased the mass of each PAM unit from approximately 1 g to 3.4 g by embedding a metal cube, and constructed a striker with a matching mass. This allowed for more efficient momentum transfer and produced a sharper, more impulse-like excitation. The added mass also helps to increase the width of the soliton (more mass, lower frequency), which allows for sharp input impulses of some low but finite width to be approximately sharp relative to the soliton's width. The striker itself was designed as a small swinging hammer that collides with the center of the striker box, with its velocity controlled by the release angle and, when needed, an added elastic band to apply acceleration.

Results

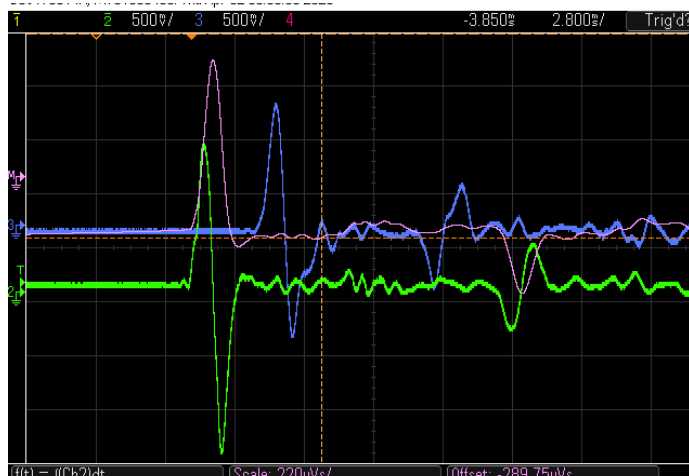


Fig. 5: Mechanical pulse resulting from the upgraded setup, where the green and blue signals are accelerometer readings from different places in the chain and the purple signal is the green accelerometer signal integrated in time to give the velocity peak. The signal shows a very large pulse in acceleration with very little wave-like response following it, as it mostly maintains its form after traveling from one sensor to the other; this agrees very well with the predicted qualities of a mechanical soliton.

With all these modifications in place (compliant boundary conditions, on-axis force application, and mass matching) we observed a qualitatively different response compared to earlier trials. The signal shown in Fig. 5 exhibits a strongly localized pulse that maintains its shape as it propagates, resembling a particle-like excitation rather than a dispersive wave packet.

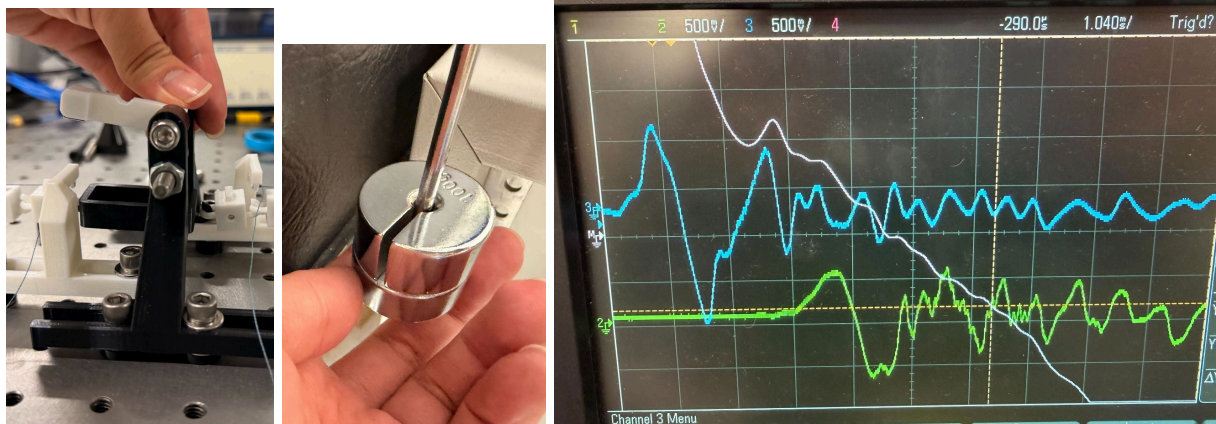


Fig 6: Without elastic band actuation, the chain was lightly struck by the falling striker while the pretension was increased to 200 grams * g, which resulted in a dispersive wave-like actuation.

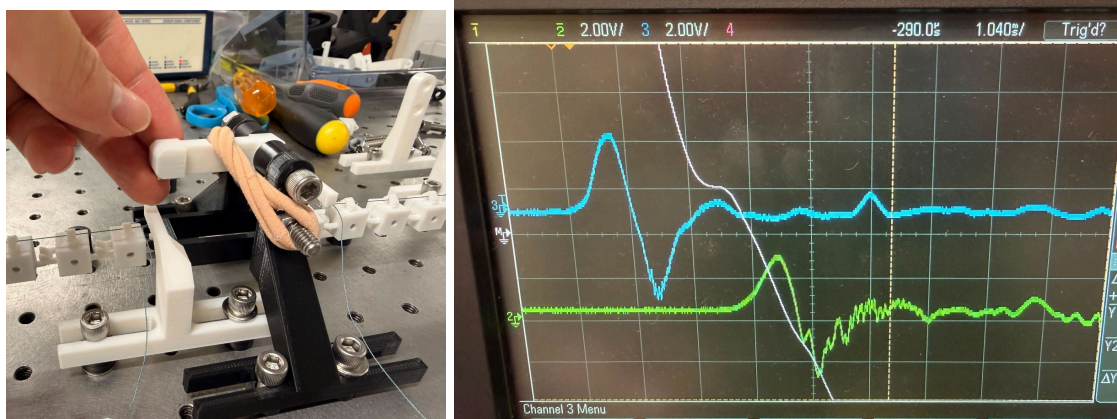


Fig 7: With the striker accelerated with an elastic band (once again increasing the striking force vs pretension ratio), we see a soliton-like response again. This verifies that the large, particle-like pulse we observed behaves very closely to a theoretically predicted soliton.

To verify that this behavior corresponds to the predicted soliton, we changed the ratio between pretension and impulse strength. When the pretension was increased and the applied impulse was relatively weak, we observed dispersive waves, consistent with the predicted behavior. When the pretension kept high but the striker velocity was increased even more (using an elastic band) the system produced stable, localized pulses. This transition from dispersive waves to localized excitations is consistent with theoretical predictions, which provides strong evidence that solitons have been successfully generated in these 3D printing-based PAM chains.

As such, we successfully established a reliable experimental platform for studying nonlinear wave propagation in polycatenated architected materials. With soliton generation now achieved, future work will focus on extending these studies to more complex geometries, including chain junctions and two-dimensional chainmail networks. In these systems, we aim to investigate how solitons interact, split, and reflect, and how their amplitudes evolve in more complex topologies.

Our immediate goal is to further characterize the soliton behavior in these 1D chains by exploiting the modularity of this experimental setup. Examples might be observing how solitons collide past each other (after 3D-printing more actuation sites and putting them against each other), how its velocity may be dependent on amplitude, or how the soliton pulse width changes in a diatomic model, where every other unit has its metal cube removed (this makes an alternating chain of heavy and light units where a heavy + light unit pair can be thought of as one large unit).